CATERPILLAR®

DEALER BEST PRACTICE SERIES

Detecting Combustion Gas Leaks in the Cooling System

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

The objective of this Best Practice initiative is to correctly diagnose combustion gases in the cooling system in a diesel engine. The principal benefit is time reduction in the repair process that allows downtime to be lowered when this situation arises. This tool must be used concurrently with the Electronic Technician.

2.0 Best Practice Description

Diagnostic tool to detect combustion gases leaking into the engine cooling system. Finning has created their own special tool design for bubble view in coolant system when a failure is present in engine series 3500B (coolant consumption and relief valve opened) nicknamed the "Burbujómetro" or "Bubble Meter".

3.0 Implementation Steps

3.1 Utilization procedure

If a combustion gas leak to the cooling system is verified, apply the next procedure.

⚠ ADVERTENCIA

El vapor de agua y el refrigerante caliente pueden causar quemaduras graves.

No afloje la tapa de llenado ni la tapa de presión de un motor caliente.

Permita que el motor se enfríe antes de quitar la tapa de llenado o la tapa de presión.

WARNING

STEAM AND HOT COOLANT CAN PRODUCE SEVERE BURNS.

DO NOT LOOSEN THE FILL COVER OR PRESSURE COVER OF A HOT ENGINE.

ALLOW THE ENGINE TO COOL BEFORE REMOVING THE FILL COVER OR PRESSURE COVER.

1. Loosen and remove the fill covers from the piston skirt side to facilitate the coolant discharge.

2.- Conectar línea entre la válvula de paso de refrigerante y el depósito

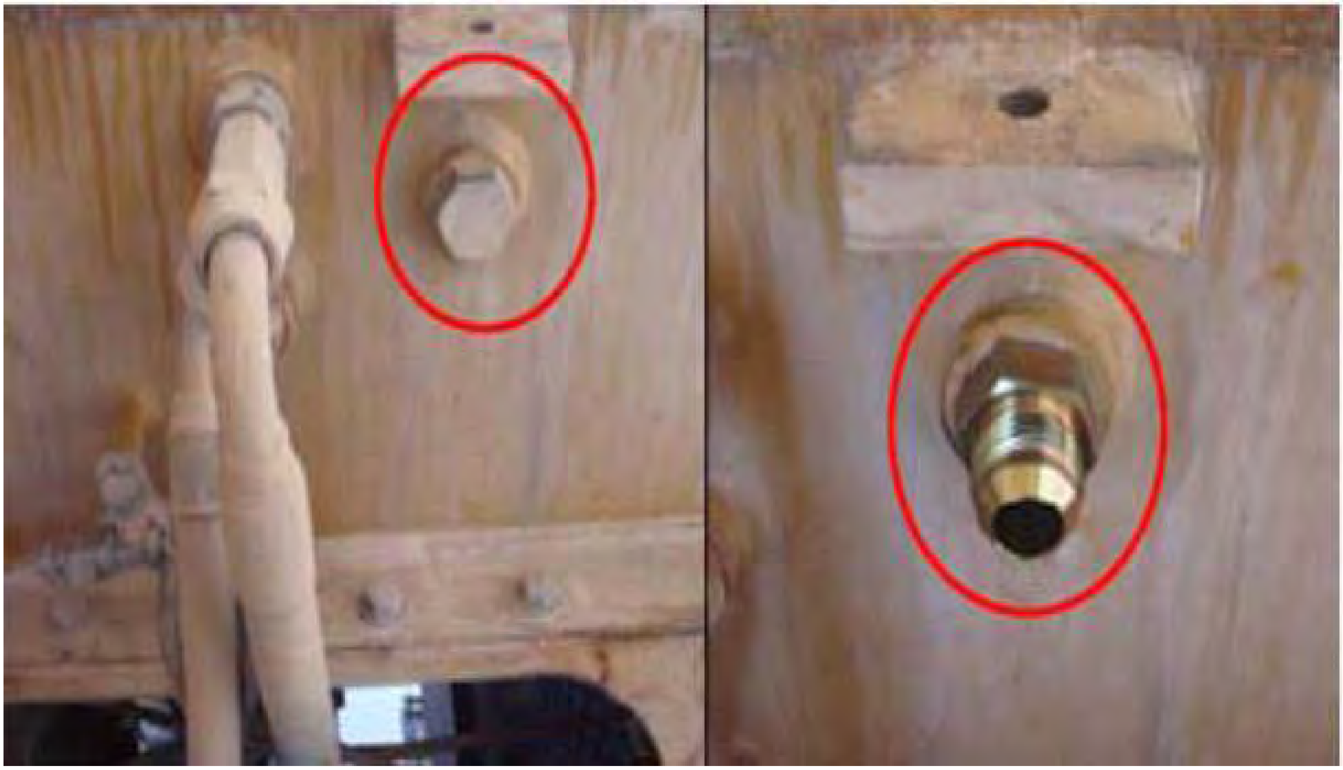
2. Connect the line between the coolant by-pass valve and reservoir.

3.- Abrir válvula de paso para que refrigerante sea descargado hasta el nivel máximo del deposito

3. Open by-pass valve so that the coolant discharges and fills up the reservoir.

4. Remove the plug cover located on the side of the case.
* See image, shunt tank in upper platform - right side - behind the fiber glass cover, the plug can be found in the shunt tank (right side jacket water fill cap)

5. Remove the plug identified in the image below to connect the return line of the test tool.



6. Behind thermostat box, remove cover Lumber 8T-6762 (a plug-pipe mounted in the front manifold) in every diesel engine bank (right and left side).

7.- Instalar los adaptadores en la zona del tapón removido con su acople rápido correspondiente

7. Install the Adapters in the area of the removed plug with the correct quick coupler.

8.- instalar la herramienta con la llave de de paso en el extremo que se conecta al tapón removido y en posición cerrada

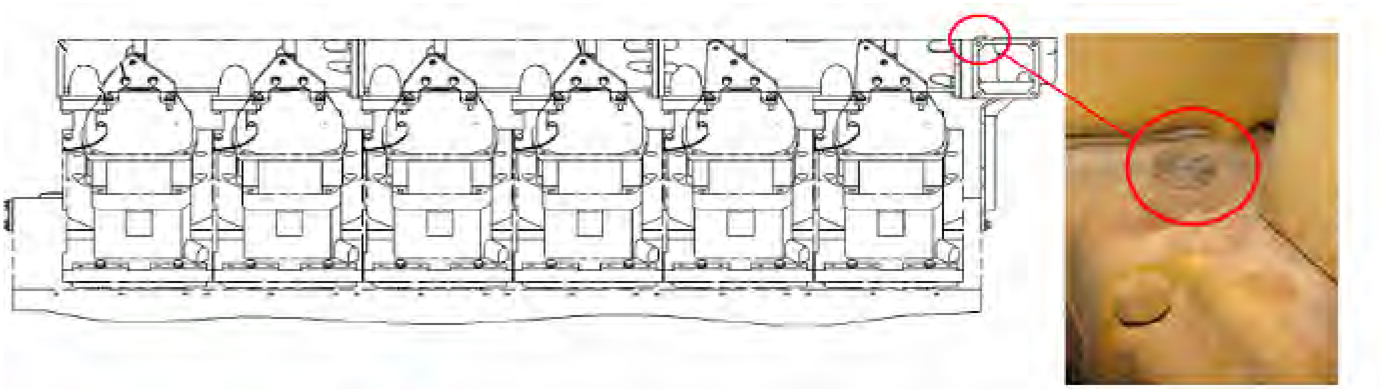
8. Install tool to the end of the by-pass valve, which connects where the removed plug existed, while in the off position.

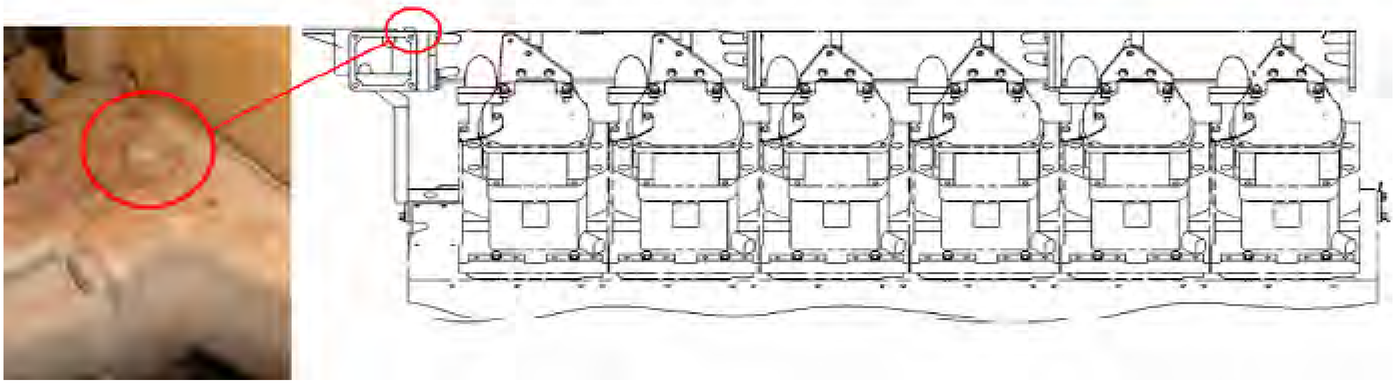
9.- Conectar la manguera de color rojo en el banco derecho a través de su acople rápido.

9. Connect red hose on the right bank with the quick coupler.

10.- conectar la manguera de color negro en el banco Izquierdo a través de su Acople rápido

10. Connect black hose on the left bank with the quick coupler.



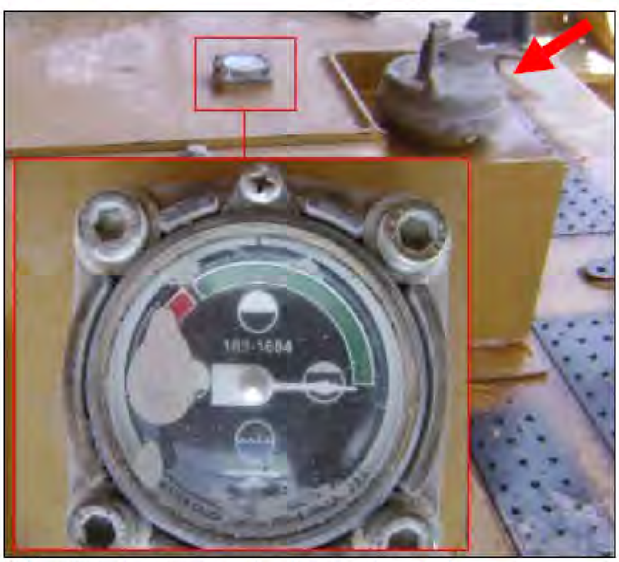


11. Connect tool between the transparent lines and the return line that must be connected to the side of the upper shunt tank (radiator cap), on the casing side.

12. Connect the end of the tool (indicated above by #4) to the side of upper shunt tank (radiator cap) to point (A), identified in Step 5.

Note: All the by-pass valves must be in the closed position, before installing the tool and before the refrigerant charge cycle.

13. Charge with the same drained refrigerant to the recommended level according to the maintenance manual.



14.- Montar la tapa de llenado dejar visible nivel de refrigerante

14. Install the fill cover ensuring coolant level visibility.

15.- Realizar la conexión del comunicador ET.

15. Connect the ET communicator.

⚠ ADVERTENCIA

Si se hace contacto con un motor en funcionamiento, se pueden sufrir quemaduras causadas por los componentes calientes del motor y lesiones personales causadas por los componentes giratorios.

Cuando trabaje en un motor que está funcionando evite hacer contacto con los componentes calientes o giratorios.

WARNING

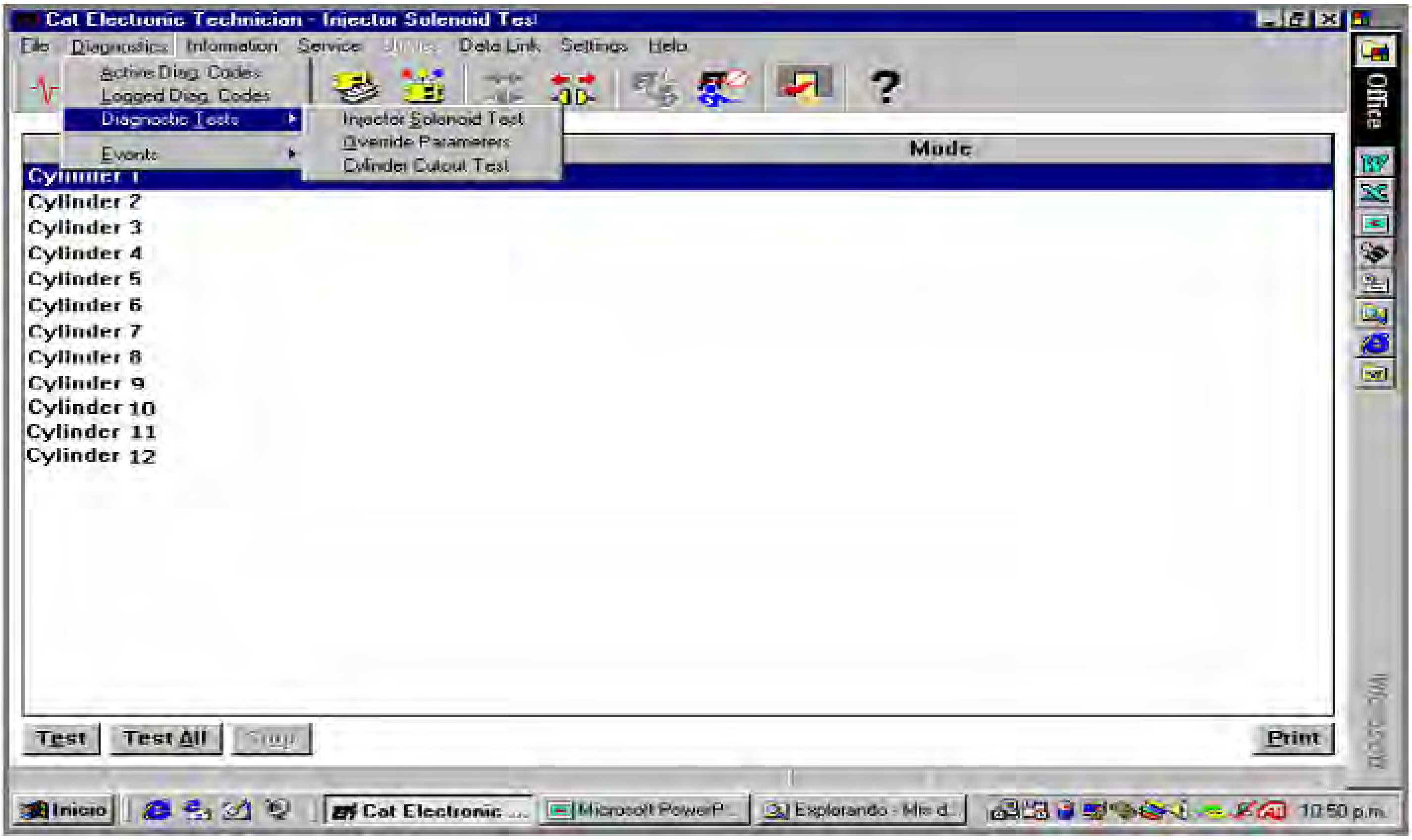
IF YOU MAKE CONTACT WITH A RUNNING ENGINE, YOU MAY SUFFER BURNS CAUSED BY THE HOT ENGINE COMPONENTS AND PERSONAL INJURY BY THE ROTATING COMPONENTS.

WHEN WORKING ON A RUNNING ENGINE AVOID THE CONTACT WITH THE HOT OR ROTATING COMPONENTS.

21. - If during a test a problem is identified. If during a test, a problem is identified, correct it. If not then continue.

22. - Open the 3 installed tool by-pass valves.

Note: The by-pass valve must be located in the described position, because if tool damage exists in the transparent zone there is a chance of a hot coolant leak, which can cause human and environmental damage.



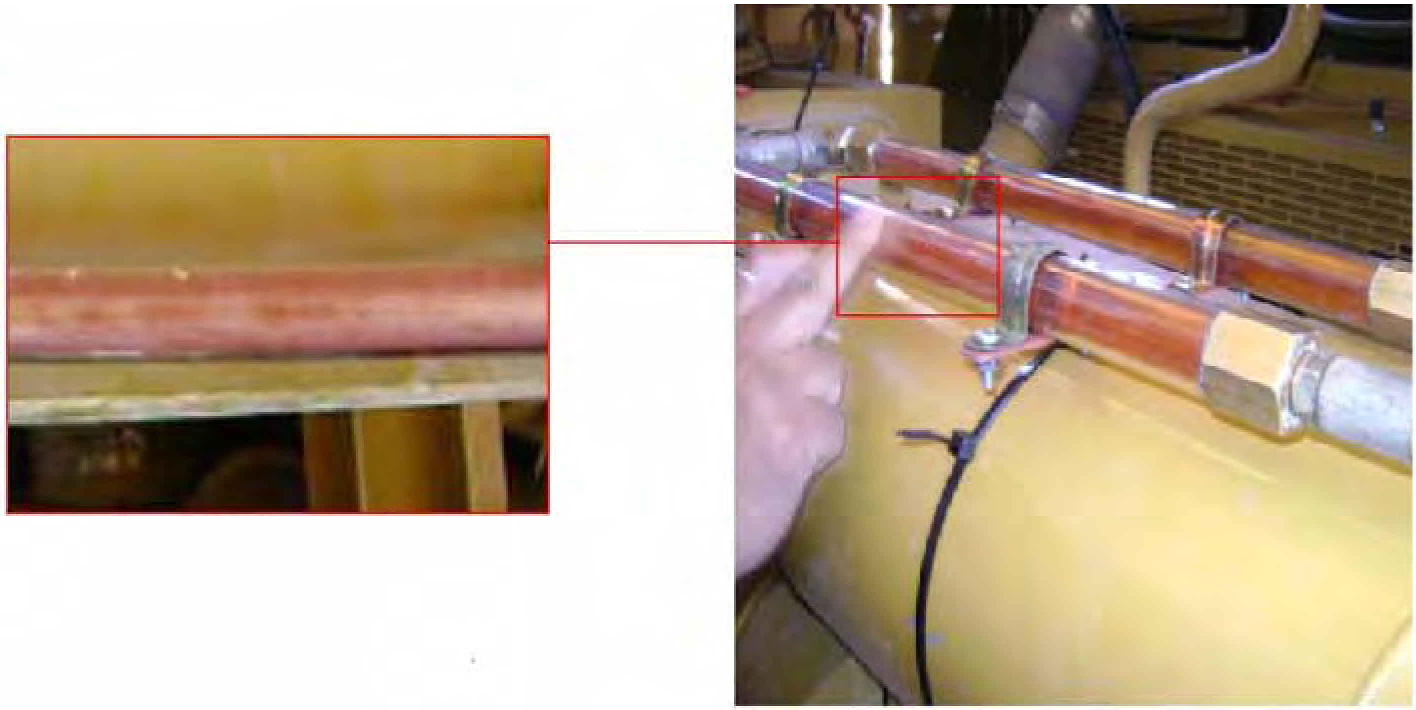
23. Increase the engine RPM to 1300 RPM without load (neutral position).

24. Select "Cylinder cutout test" on the Diagnostic menu, Diagnostic test sub menu, on the ET tool bar.

25. Cut all the right side of the rear and front module, from the unit 1 (front engine) to unit 11 (rear engine) and proceed to activate each individual injector solenoid for 30 seconds.

Note: Always must be an active unit from the evaluated bank.

26. When the cylinder cut-out is done and after activation, you must check for combustion gas bubbles and pressure generation. These bubbles are visible through the transparent hose indicating each engine side, right and left on the thermostat box.



27. The picture above is evaluating the right side of a 3524B engine so you must pay attention for bubbles in the transparent hoses coming from the right side.

28. Apply the same procedure for the left side to identify the affected unit.

29. To check the diagnostic, use the Dummy Injector procedure on the unit with problems.

30. Once the power unit with problems is identified, proceed with the unit repairs.

4.0 Benefits

- Increased safety. There isn't contact between technician and nitrogen.
- REDO reduction.
- Diagnostic repair time reduction.
- **Avoid all engine regulation, decreasing probabilities of failures in good units.**

5.0 Resources Required

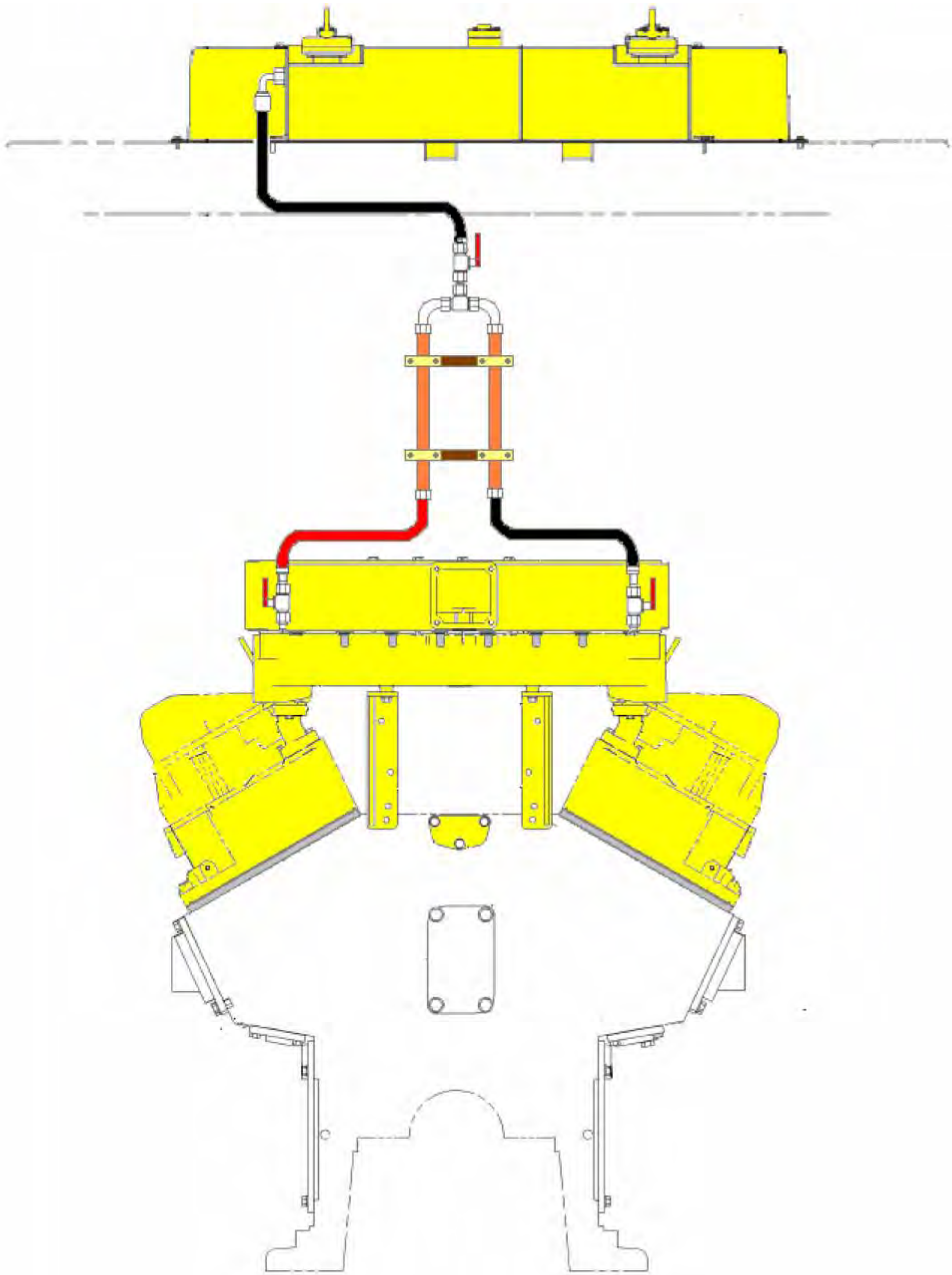
- 1 polycarbonate tube (50 - 60 cm) ¾” internal diameter.
- 1 polycarbonate tube (50 - 60 cm) 1” internal diameter.
- Hoses with couplings.
- Adapters, by-pass valves and one base.
- 2 suitcases to store and protect these tools.

6.0 Supporting Attachments / References

1.- Visor
2.- Mangueras identificación de bancos
3.- Llaves de paso
4.- Mangueras bote superior
5.- Acople rápidos

1.- Viewfinder.
2.- Banks identification hoses.
3.- Pass valves.
4.- Upper boat hoses.
5.- Quick couplings.

1. Viewfinder
2. Bank Identification Hoses
3. By-Pass Hoses
4. Upper Shunt Tank Hoses
5. Quick Couplers



7.0 Related Best Practices

While this Best Practice highlights Finning's way of testing for gas leaks, below is Caterpillar's documented process to check for a gas leak in a combustion engine. It is left to the dealer to decide which practice is best for them.

<u>CATERPILLAR Method</u>

1. Remove rocker arm assembly.
2. Remove Injector.
3. Make sure all Fuel is out of the cylinder.
4. Install "Dummy Injector" by using injector hold down clamp.
5. Pressurize "Dummy Injector" to a maximum of 100 psi (compressed air or nitrogen).
   a. Warning: When pressurizing the cylinder, the engine may rotate.
   b. Warning: DO NOT exceed 100 psi when using this tool.
6. Open ball valve until the gauge stabilizes at 100 psi.
7. Close ball valve.
8. Watch gauge for slow and steady pressure release back to zero.
9. If gauge drops to "Zero" in less than 2 seconds, the cylinder is suspect.
10. Repeat steps 1-9 for each additional cylinder head.

8.0 Acknowledgements

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